Adenosine

Scope of Practice	P
Classification	• Antiarrhythmic
Usual Presentation	• 6mg/2ml vial
Mechanism of Action	• Slows impulse formation in the sinoatrial (SA) node • Slows conduction time through the atrioventricular (AV) node • Can interrupt reentry pathways through the AV node. • Onset Time: Immediate / Peak: 20seconds / Duration: 40seconds
Indications	• Chemical Cardioversion of Narrow complex Supraventricular Tachycardias (SVT) and regular monomorphic Ventricular Tachycardias
Contraindications	• Known allergy • Wolf Parkinson White Syndrome with Atrial Fibrillation • Known Sick Sinus Syndrome (without a pacemaker in place) • Previous 2nd or 3rd degree heart block (without a pacemaker in place)
Precautions	• Asthma or Severe Reactive Airway Disease (COPD)
Side Effects	• Bradycardia / Asystole (usually transient) / VF/VT • Hypotension • Difficulty of breathing • Chest Pain • Facial Flushing
Adult Dose & Administration	• **1st Dose: 6mg** rapid IV/IO push over 1-2 seconds • **2nd Dose: 12mg** rapid IV/IO push over 1-2 seconds • Each dose should be followed by a 10-20ml Normal Saline flush. • Consider elevation of the arm
Pediatric Dose & Administration	• **1st Dose: 0.1mg/kg** (maximum 6mg) rapid IV/IO push over 1-2 seconds • **2nd Dose: 0.2mg/kg** (maximum 12mg) rapid IV/IO push over 1-2 seconds
Use in Pregnancy and Lactation	• Pregnancy: Fetal risk cannot be ruled out • Lactation: Fetal risk cannot be ruled out • Safety has not been demonstrated, however if indicated, the balance of risk is in generally favor of administration.
Additional Information	• At the time of conversion to normal sinus rhythm, a variety of new rhythms may appear on ECG including: • Short period of asystole, premature ventricular contractions, atrial premature contractions, sinus bradycardia, sinus tachycardia, skipped beats, AV nodal block.